

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — Experienced Manager

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Summary

Data Scientist & Machine Learning Engineer with strong leadership in machine learning and statistical modeling who generated over \$1.3 billion in business value. Specializations include design of data solutions, implementation of MLOps pipelines on AWS and Snowflake, enhancing customer engagement.

Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ — 09/2022 to Present

- Architected and developed end-to-end data products using solid programming skills, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Performed data mining on structured, semi-structured, and unstructured data to assemble a training dataset from multiple sources, considering thousands of parameters derived from dealer and product information
- Used LLMs and natural language processing to find similarities in service jobs and performed feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Utilized Python and Sagemaker to train AI models, leveraging machine learning algorithms such as neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and led to more accurate predictions of equipment repair times
- Engineered production-ready MLOps pipelines on AWS and Snowflake, establishing best practices for cloud architecture. Managed project lifecycles, automated enterprise networks, and implemented monitoring strategies to assess model performance and business outcomes.
- Implemented robust information management strategies, such as STAR schema, to ensure data integrity and accessibility for the Snowflake data lake, improving data reliability and supporting informed decision-making

Reliability Analytics Manager — Amazon.com — Tempe, AZ — 02/2022 to 09/2022

- Applied advanced analytics and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis and wrote an influential white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Utilized advanced data science techniques to extract actionable insights, driving strategic decision-making and improving operational outcomes
- Designed and optimized regression and recommendation systems, achieving two-time champion status in Amazon Machine Learning University competitions, which improved predictive accuracy and recommendation quality in practical applications
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager — Empire-Cat — Mesa, AZ — 01/2012 to 02/2022

- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days
- Developed an time-series MLOps pipeline for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting
- Supervised 8 engineers and acted as a customer-facing expert on mining equipment. Implemented a data-driven failure analysis program, boosting an aging fleet's physical availability to over 90%.
- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding \$1.2 billion in savings over 6 years as Reliability Engineering Manager prior to promotion
- Leadership experience with business acumen to build trust and enhance customer experience and relationships to promote data-based decisions. Partnered on design of experiments to improve reliability, safety and durability

Certifications

- **Amazon Machine Learning University:** Tabular Data with Autogluon
- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX - edX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **CSCMP:** Supply Chain Foundations Demand Planning Professional Certificate
- **Data Camp:** AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate
- **Empire-Cat:** Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** RAG and Agentic AI Professional Certificate, Deep Learning & Neural Networks with Keras
- **Komatsu North America:** Certified Technical Communicator
- **LinkedIn Learning:** Data Engineering with dbt
- **Noria:** Oil Analysis I, II, III
- **Society of Automotive Engineers:** Cost and Finance for Engineers
- **Udemy:** Streamlit for Snowflake Masterclass

Education

Bachelor of Science in Mechanical Engineering – Arizona State University — Tempe, AZ — 06/2003 to 05/2006

Hard Skills

A/B Testing, Agile Methodology, AI/ML, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Git, Linear Algebra, LLM, Mathematics, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Sagemaker, Scikit-learn, Sagemaker, Scripting Language, SDLC, Simulation, Snowflake, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn